

1. human ROS1 BINDING "ligand: ATP" (9 aa) and worm Q11112

TOOLS: kmerseek sdm12 k6 (chosen for BINDING in Stage 0) and protein20 k5; phmmer, MMseqs2, MMseqs2 iterative, Foldseek, ProstT5, Reseek; Kyte-Doolittle scan

- kmerseek: call lands on the feature (>= 80% of call inside, >= 30% covered)

call mostly inside the feature, lower IoU than kmerseek

call overlaps the feature but < 50% of the call is inside it

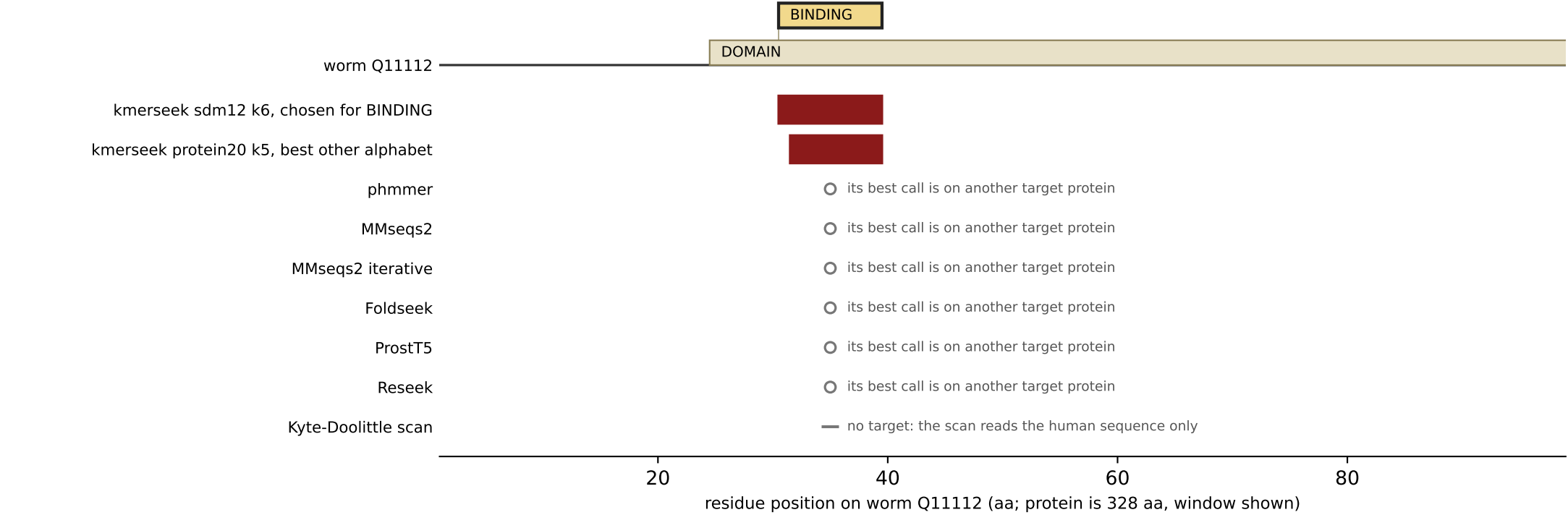
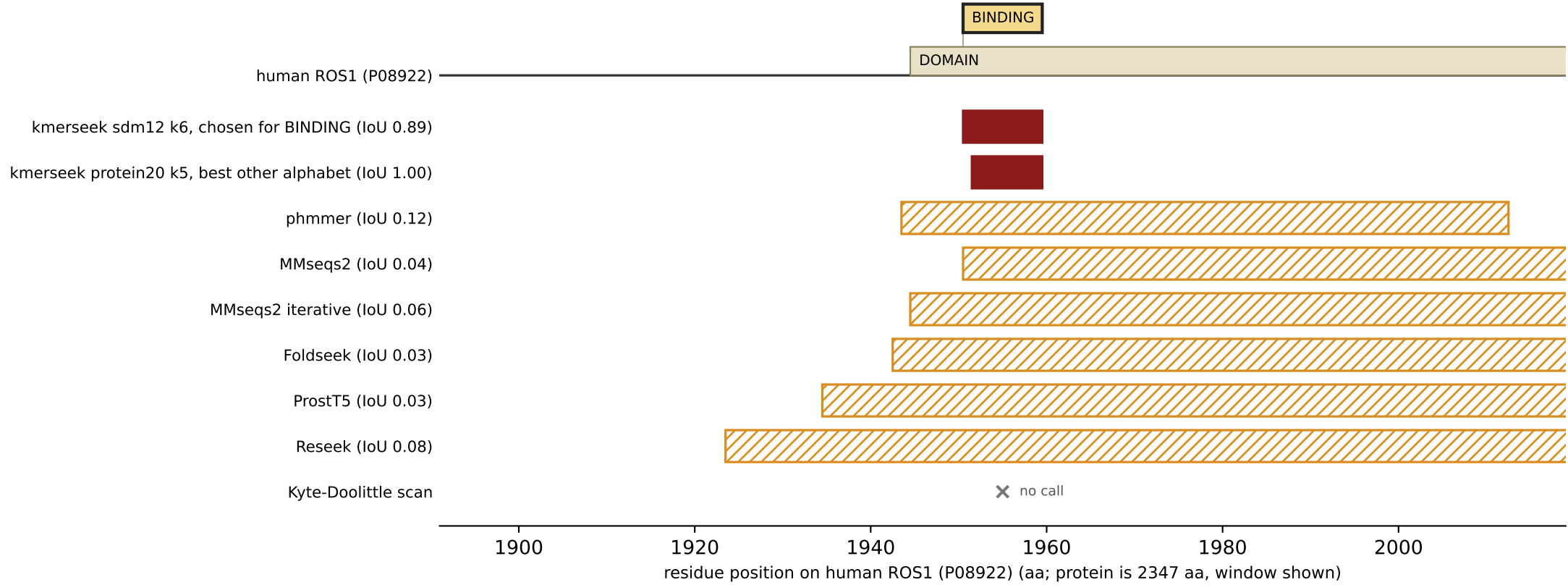
call reaches kmerseek's IoU or higher

no call of this feature type overlaps the feature
- the tool's best call is on another target protein

Kyte-Doolittle scan: no target protein, it searches nothing

the Swiss-Prot feature being scored (human) / transferred from (target)

other Swiss-Prot features, labelled by type



Hypothesis: kmerseek places the human BINDING feature "ligand: ATP" on its worm counterpart and the structure tools do not.
Conclusion: kmerseek IoU 0.89, 8 of 9 residues identical; Foldseek spills; ProstT5 spills; Reseek spills; phmmer spills; MMseqs2 spills; MMseqs2 iterative spills; Kyte-Doolittle scan did not land; pLDDT 67, disorder fraction 0.0.